

Plan B

De-specializing SALMONN-SQA into a calibrated, multi-dimensional speech-quality rater

Turning a lenient noise specialist into a model that perceives, names, and scores every common degradation — and emits a de-compressed, calibrated MOS.

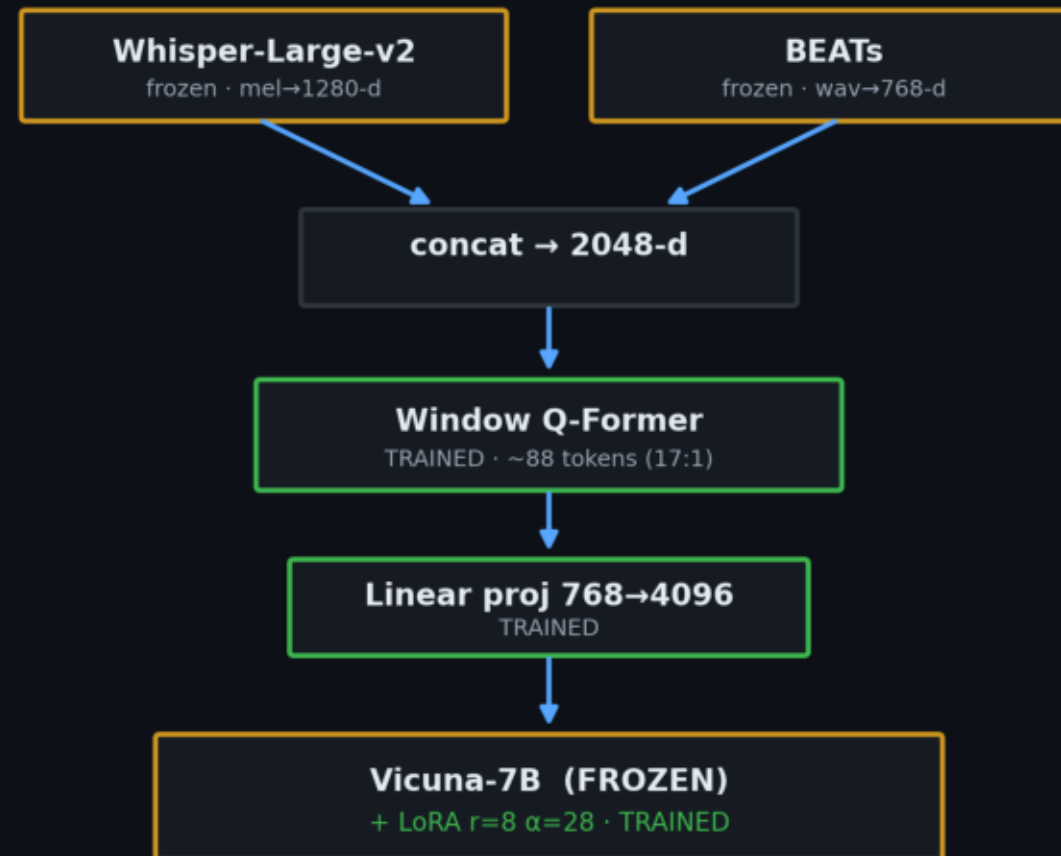
Clement Laroche · final model ckpt_stage2_v3 · LibriTTS-R → VoiceBank-DEMAND held-out

The model: SALMONN-7B, SQA-finetuned

Audio LLM: two frozen encoders → a trained audio bottleneck → frozen Vicuna-7B + small LoRA.

- Whisper-Large-v2 — mel→1280-d. ASR-trained, deliberately reverb-invariant.
- BEATs — raw waveform → 768-d acoustic-event features.
- Window Q-Former — ~0.33s windows → ~88 tokens. TRAINED. ~17:1 compression.
- Linear projection 768→4096, spliced at <SpeechHere>. TRAINED.
- Vicuna-7B-v1.5 + LoRA ($r=8$, $\alpha=28$). Base FROZEN, LoRA TRAINED.

Plan B trains only the green blocks + LoRA. Encoders stay frozen.



The problem: a lenient noise specialist

Good additive-noise descriptions — but as a quality meter, weak and blind to whole classes.

① MOS isn't a usable scale

- Coarse 5 discrete values, floored at 2.5. Agrees with PESQ/NISQA/DNSMOS at only ρ 0.40–0.49
 - — yet those metrics agree with each other at 0.72–0.82. It is the outlier.

② Blind to whole degradation types

- Reverb $\rho(\text{MOS}, \text{severity}) = -0.11$ (no response). Weak on bandwidth (-0.39), clipping (-0.33).

③ Blind to enhancement

- Denoise: every objective metric improves, MOS moves $+0.03$.

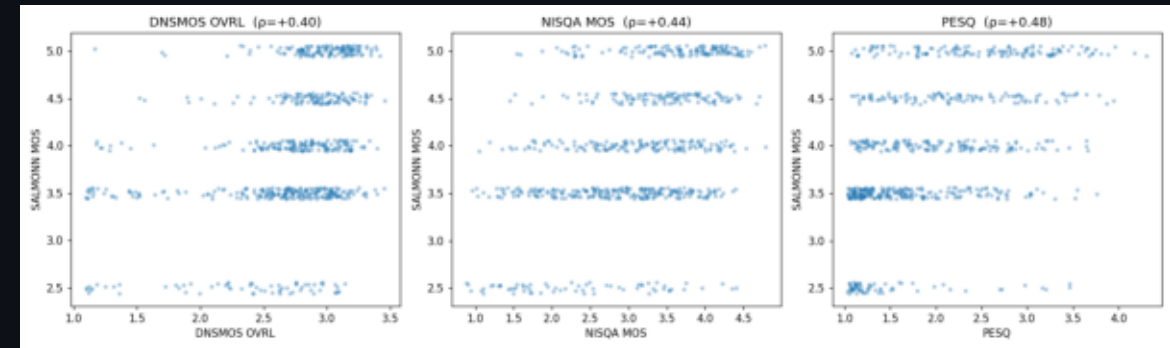
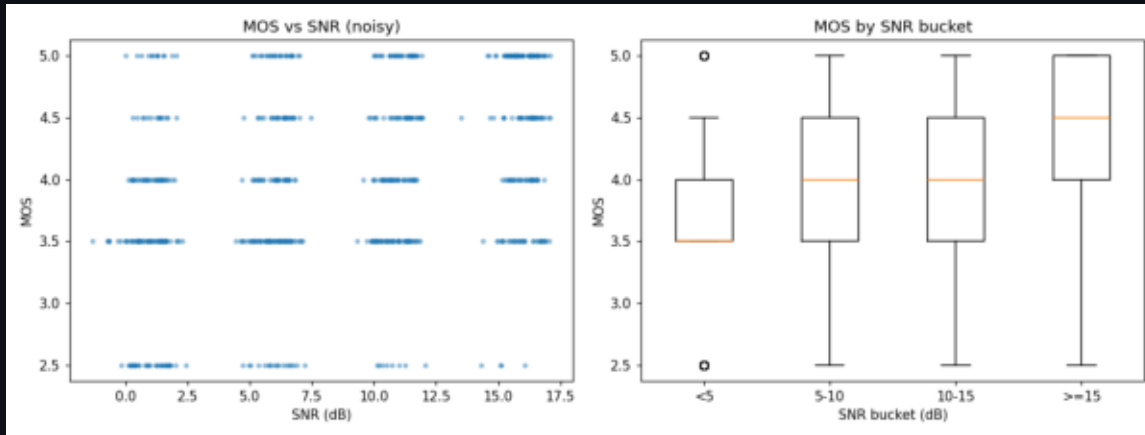
④ Brittle output

- OOD degradation $\rightarrow$ made-up JSON schema. Acquiescent: leading yes/no $\rightarrow$ always YES.

Goal: keep the descriptive ability; add perception + calibration across ALL degradations.

Evidence of the blindness (original model)

- Left: MOS barely tracks SNR, and sits in a narrow high band — the floored, compressed scale.
- Right: vs DNSMOS / NISQA / PESQ the MOS quantizes into a few horizontal stripes (ρ 0.40–0.49).



Experiment design: de-risk, then train



Ask the key question cheaply, first

- Whisper is reverb-invariant; Q-Former compresses 17:1. If reverb/bandwidth info is destroyed
 - in the frozen front-end, no LoRA training recovers it → would need to unfreeze encoders (heavy).

The hypothesis

- Blindness lives in the LLM's learned read-out (only ever trained on noise), not the architecture.

If true, the fix is data + targets — not surgery on the encoders.

Stage 0 — is degradation decodable from frozen features?

Known-parameter degradations → features at 4 frozen

taps → linear probe (PCA+ridge),

scored by R^2 +Spearman, held out by utterance.

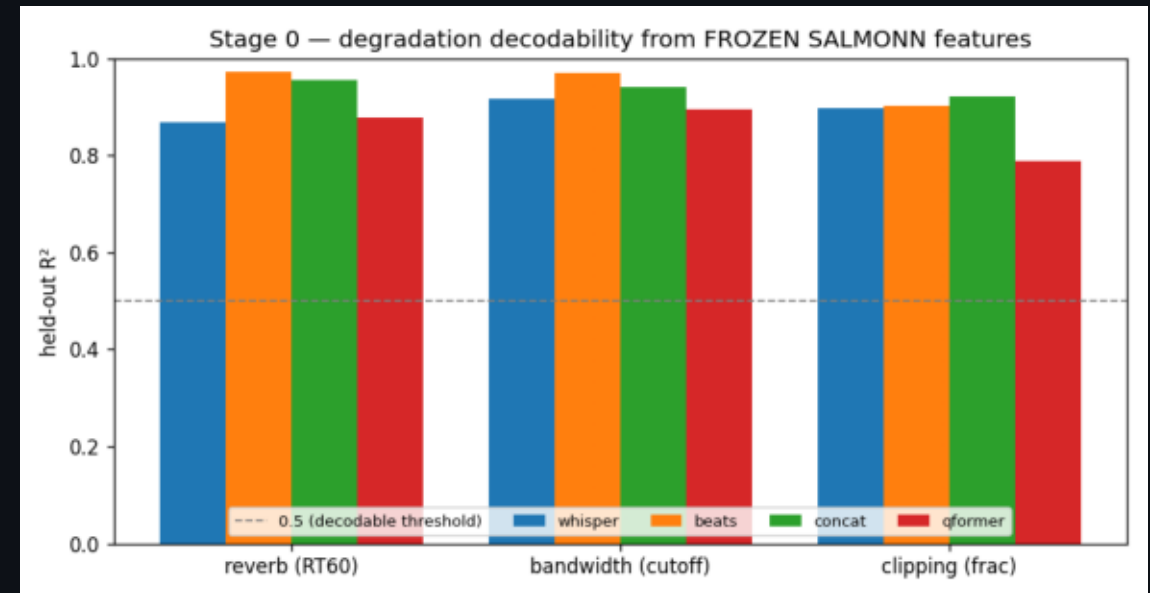
A linear probe is the point: tests whether info is linearly

decodable, not buried.

Taps: whisper · beats · concat · qformer (the bottleneck that reaches the LLM)

Result — PASSED. R^2 0.82-0.95, Spearman 0.90-0.98 at every tap, incl. Q-Former output.

- Decision: keep encoders frozen; the fix is data + targets.



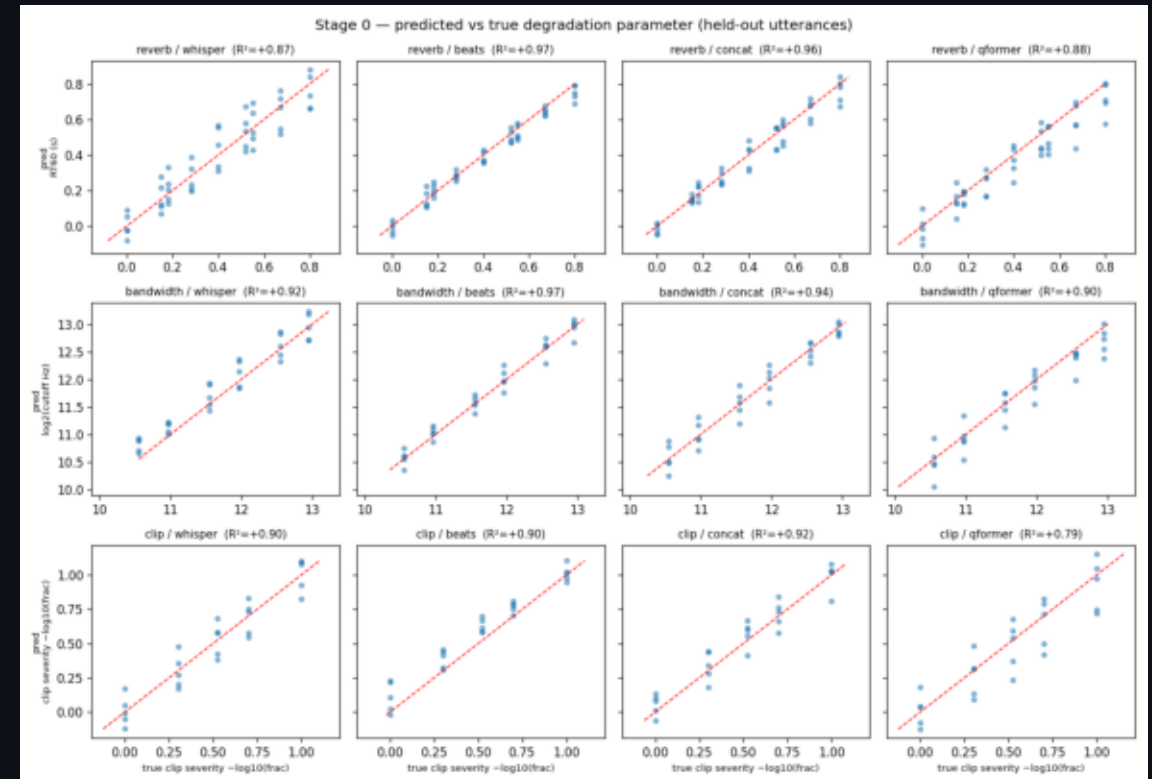
Stage 0 — the signal survives to the LLM's doorstep

Where the blindness actually is

- Predicted-vs-true RT60 / cutoff / clip-fraction sit on $y=x$ at every tap, incl. qformer.
- The degradation reaches the LLM intact — the model was never taught to read it
 - (its read-out head was trained only on additive noise).

Each axis is probed one-at-a-time → proves per-axis detectability, not type-discrimination.

Discriminating WHICH degradation is the residual that v3's synthetic-type data fixes.



Data generation: on-the-fly, known-parameter corpus

Clean speech — leakage-safe

- Train: LibriTTS-R train-clean-100 — 4,000 clips. Val: dev-clean — 300 clips.
- No VCTK speakers → disjoint from VoiceBank-DEMAND eval. No DEMAND noise in training.

Description text — Opus 4.8 paraphrase

- ~740 unique profiles → one Opus 4.8 call each → grounding-verified prose
- All 4,300 descriptions paraphrased, 0 template fallbacks.

axis	how	train
reverb	real measured RIRs + synth exp-decay	2,783
noise	real MUSAN + synth white/pink/brown	512
codec	real Opus/MP3 → bandwidth	509
bandwidth	Butterworth low-pass	DSP
clipping / disc.	hard-clip / frame-drop	DSP
loudness	re-gain	DSP
clean	untouched	373

real + synthetic mix is the v3 addition — closes the train/eval gap (eval sweep uses synthetic types).

Choice of artefacts & the severity map

8 degradation classes, each scored 1–5 from its exact applied parameter (5=pristine,1=severe).

Three design choices

- Perceptual bands — thresholds follow speech-quality convention (telephone BW $\approx$ 3.4 kHz).
- Reverb floors at 4 for any applied RIR — measured RIRs measured at distance, never truly dry;
 - very low DRR (<-15 dB) drops one more band. (Fixed an early RT60-0.15-scored-clean mislabel.)
- Use the measured effect, not the knob — clip/disc. on measured fraction; codec→measured rolloff.

axis	parameter → score bands
noise	SNR: $\geq 30 \rightarrow 5$, $20-30 \rightarrow 4$, $12-20 \rightarrow 3$, $6-12 \rightarrow 2$, $< 6 \rightarrow 1$
reverb	RT60: $< .25 \rightarrow 4$, $.45 \rightarrow 3$, $.70 \rightarrow 2$, $> .70 \rightarrow 1$; DRR $< -15 \rightarrow -1$
bandwidth	cutoff: $\geq 7500 \rightarrow 5$, $6000 \rightarrow 4$, $4000 \rightarrow 3$, $2500 \rightarrow 2$, $< 2500 \rightarrow 1$
clipping	frac: $0 \rightarrow 5$, $< 1\% \rightarrow 4$, $5\% \rightarrow 3$, $15\% \rightarrow 2$, $> 15\% \rightarrow 1$
discontinuity	loss: $0 \rightarrow 5$, $< 2\% \rightarrow 4$, $5\% \rightarrow 3$, $10\% \rightarrow 2$, $> 10\% \rightarrow 1$
loudness	$ \Delta \text{gain} $: $\leq 3 \rightarrow 5$, $6 \rightarrow 4$, $12 \rightarrow 3$, $20 \rightarrow 2$, $> 20 \rightarrow 1$

Targets: calibrate-then-describe

Per-dimension 1-5 scores

- From exact params — direct fix for 'quality prior overrides the named dimension'.

Overall MOS

- $0.55 \cdot \min + 0.45 \cdot \text{mean of dim scores (worst-axis-dominant)}$, blended 70/30 with fused
 - PESQ+NISQA+DNSMOS. Pure metric fusion rejected: PESQ floors on reverb, DNSMOS reverb-blind.

Expected I/O

- Input: waveform + sqa_full instruction.
- Stage 1 target: score block ONLY (learn calibrated numbers first).
- Stage 2 target: scores + paraphrased description + Overall MOS.

Cross-entropy on target tokens only; prompt + 88 audio embeds masked to -100 .

Training: two-stage LoRA-SFT

What LoRA does here

- Learn tiny low-rank adapters ($r=8$, $\alpha=28$) in Vicuna's projections — a few M params that
 - re-shape the read-out without disturbing the 7B frozen base.
- Trainable: Q-Former + speech→LLaMA proj + Vicuna-LoRA. Frozen: Whisper, BEATs, Vicuna base.

single GPU · AMP · batch 2 × accum 8 · lr 1e-5 · warmup→cosine

Why two stages

- Learn calibrated numbers in isolation first (no description tokens competing for gradient),
 - then describe + emit MOS on top of a representation that already rates correctly.

stage	target text	from	ep
1 — calibration / (sqa_score)	per-dimension score block only	released SQA ckpt	2
2 — reasoning / (sqa_full)	scores + description + Overall MOS	Stage 1 best	4

Results: all five findings re-measured

Held-out synthetic degradation sweep on VoiceBank-DEMAND clean (disjoint speakers).

#	Original finding	v3
1	MOS↔SNR ρ 0.37, lenient	ρ 0.50; penalizes low SNR; names noise 99% (low SNR)
2	ρ 0.40–0.49; 5 values floored	ρ 0.69–0.75 (in metrics' band); 81 distinct (2.10–4.89)
3	reverb -0.11 , bw -0.39 , clip -0.33	reverb -0.95 , bw -0.81 , clip -0.83 , noise -0.93
4	Enhancement MOS $+0.03$, $\rho \approx 0$	MOS gain $+0.68$ (76% of files), ρ $+0.22$
5	Acquiescence $\rightarrow$ always YES	calibrated discriminative scores; N/A

5→81

distinct MOS values

−.11→−.95

reverb ρ

0 / 108

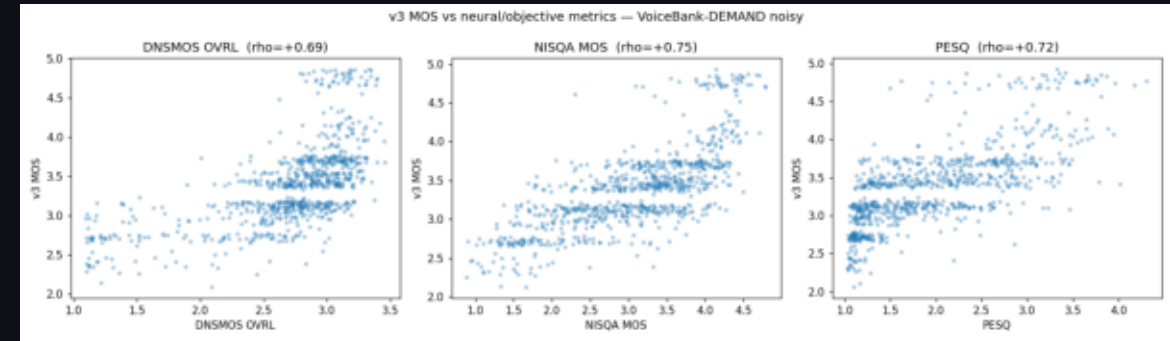
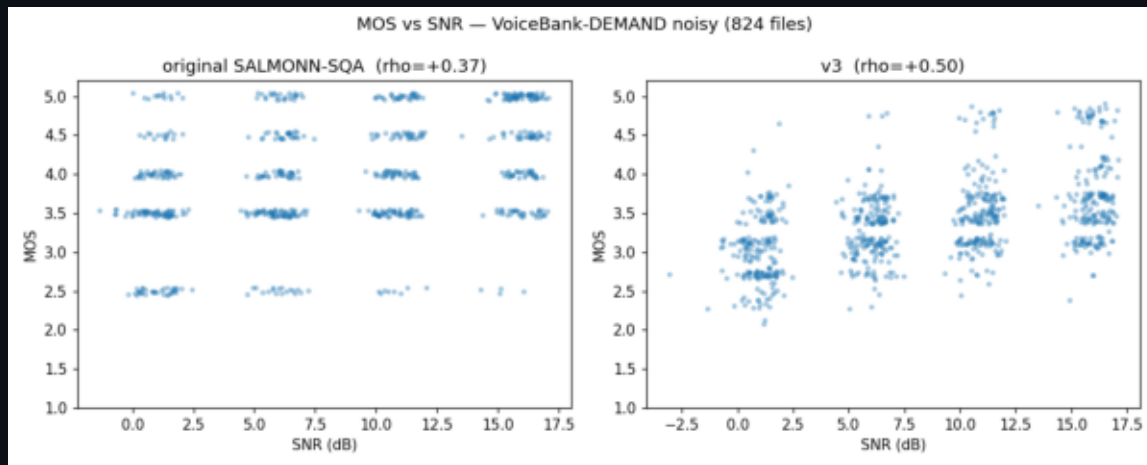
degenerate outputs

90%

real-noise naming

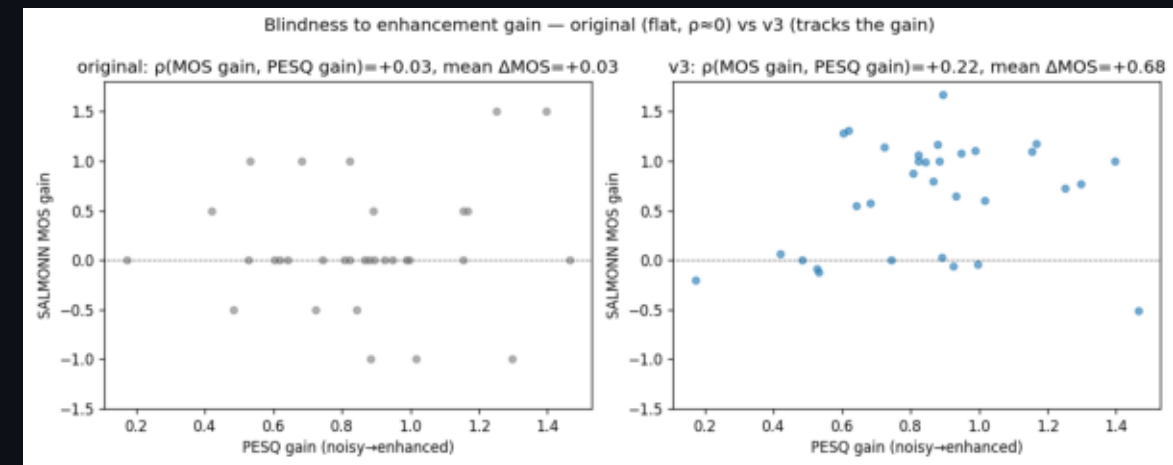
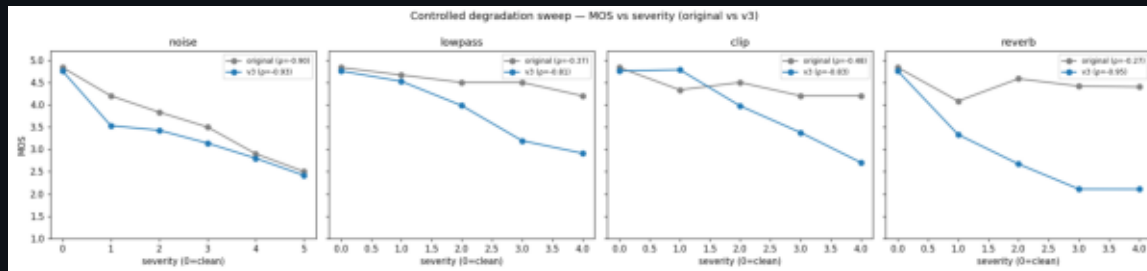
Results: the MOS scale is rebuilt

- Left: MOS vs SNR — v3 cloud spreads the full range and slopes with SNR (was a flat band).
- Right: v3 MOS vs metrics — continuous, rank-aligned (ρ 0.69–0.75); stripes gone.



Results: every axis now responds

- Left: MOS vs severity — monotone downward everywhere, incl. the former reverb blind spot.
- Right: enhancement — v3 MOS rises with denoiser gain (usable as an enhancer evaluator).

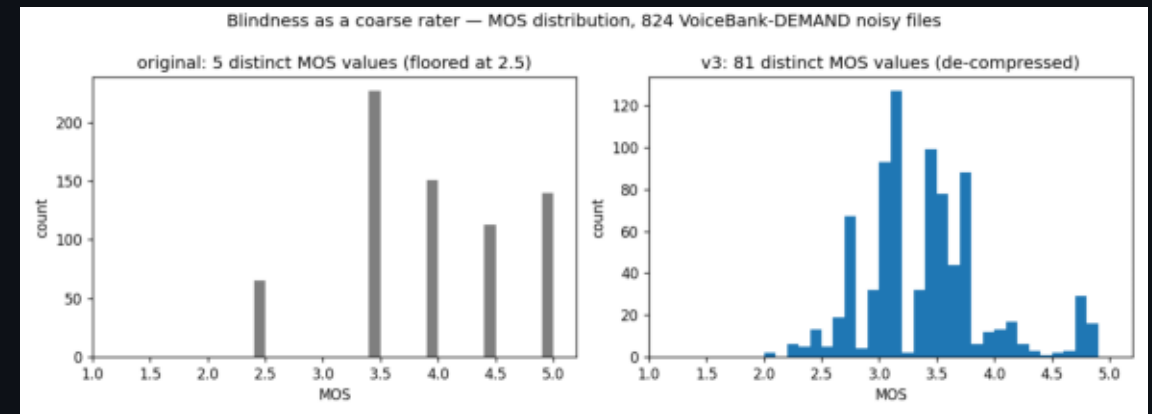


Results: from a 5-rung ladder to a continuous meter

- Original emitted a handful of MOS values, floored well above the bottom of the scale.

v3 uses 81 distinct values spanning 2.10-4.89 — it meters quality instead of bucketing it.

This de-compression lets rank-correlations climb into the band where the metrics agree.



Bottom line

v3 turns a lenient noise specialist with a weak 5-value MOS into a calibrated rater:

- MOS de-compressed (5→81), agreeing with PESQ/NISQA/DNSMOS as well as they agree with each other
- Perceives and ranks every degradation type — incl. the former reverb blind spot
- Tracks enhancement gain → usable as an automatic enhancer evaluator (it wasn't before)
- Names degradations in natural, grounded prose; 0 degenerate outputs

Per-dimension structured scores are the most reliable signal. The whole fix was data + targets —

encoders never unfrozen, validated up-front by the Stage 0 probe.

Optional further work

- Scale clean pool to train-clean-360 · more degradation types · scalar regression head if banding remains.